

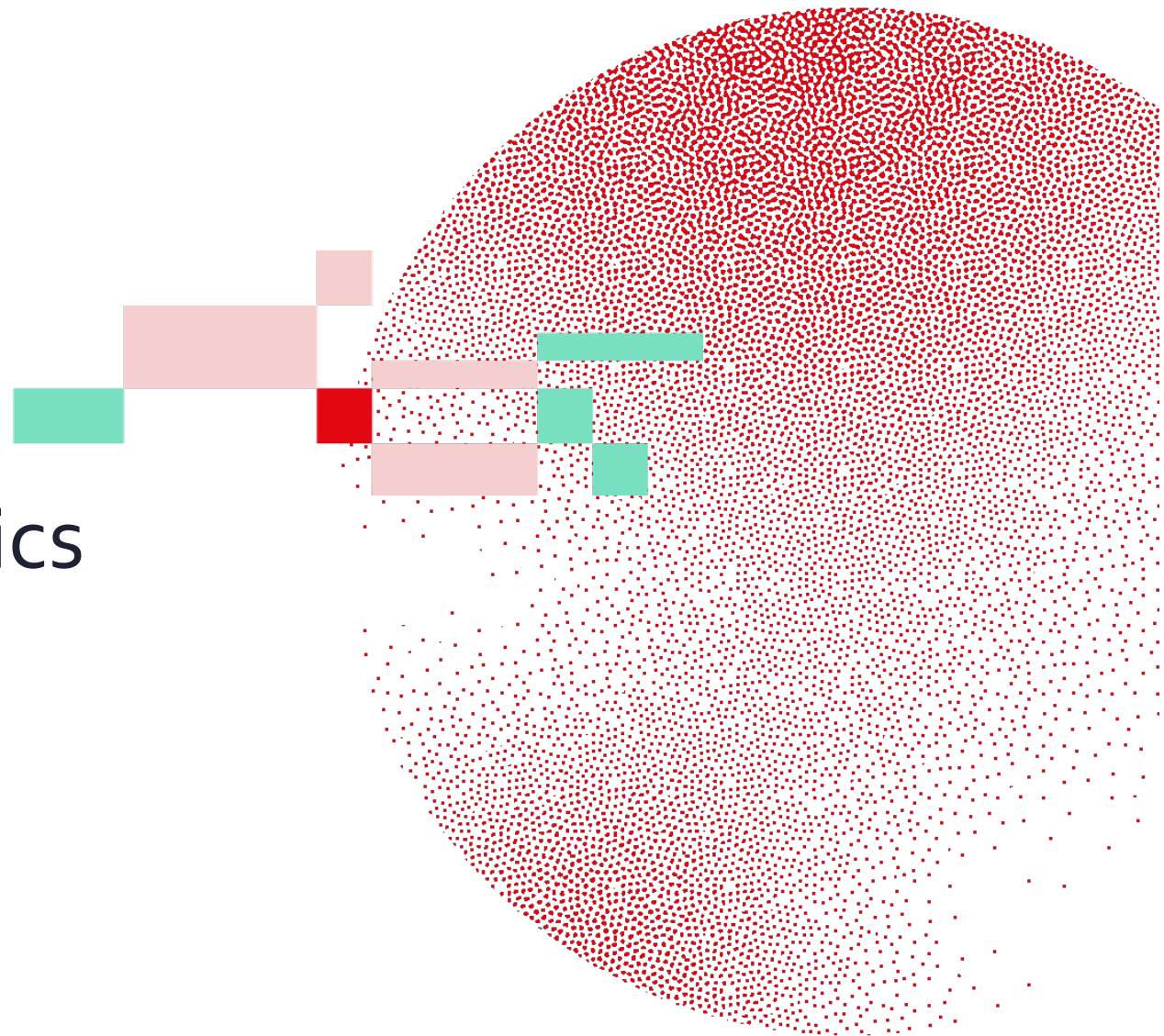


Swiss Institute of
Bioinformatics

INTRODUCTION TO BIOINFORMATICS:

Clinical Bioinformatics

V. Barbié, Director Clinical Bioinformatics
Zürich, 02 December 2025



Outline

What is
clinical bioinformatics

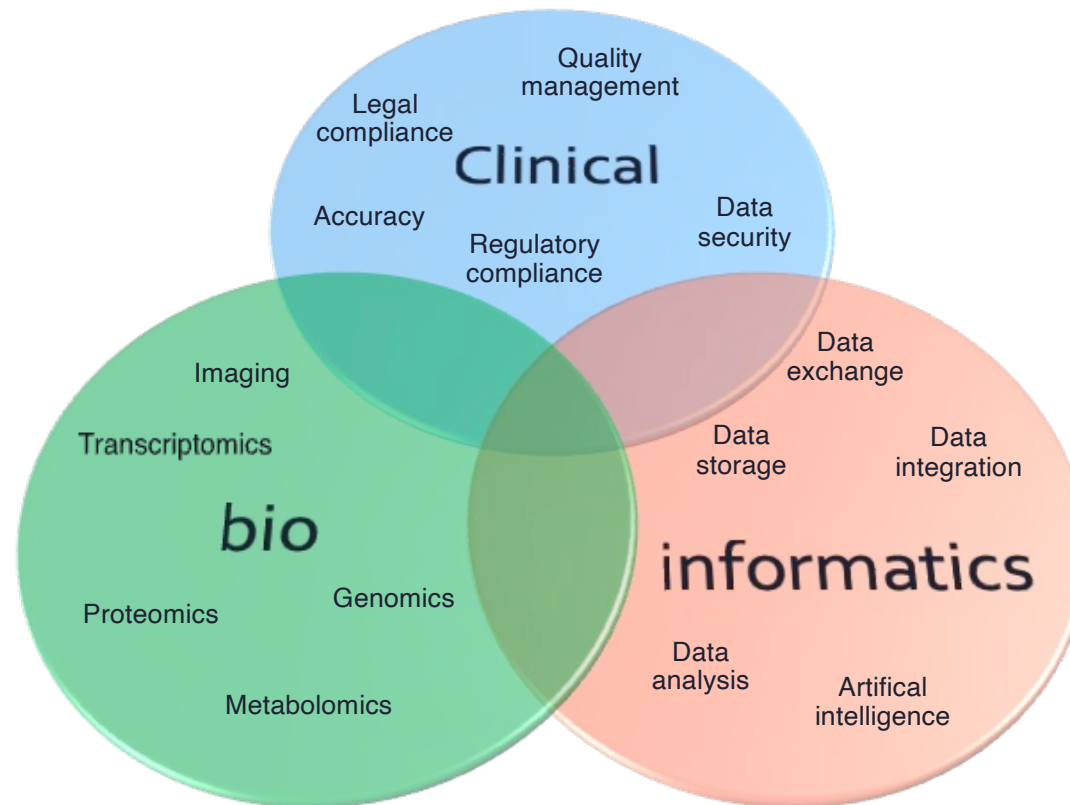
Why clinical bioinformatics?
*Next Generation Sequencing (NGS)
in medical diagnosis*

Overview of an oncology NGS
diagnostic pipeline

Other considerations



What is clinical bioinformatics?



Outline

What is
clinical bioinformatics

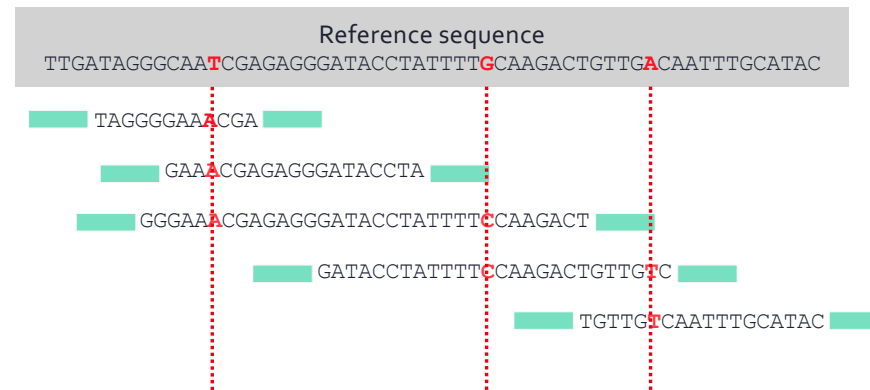
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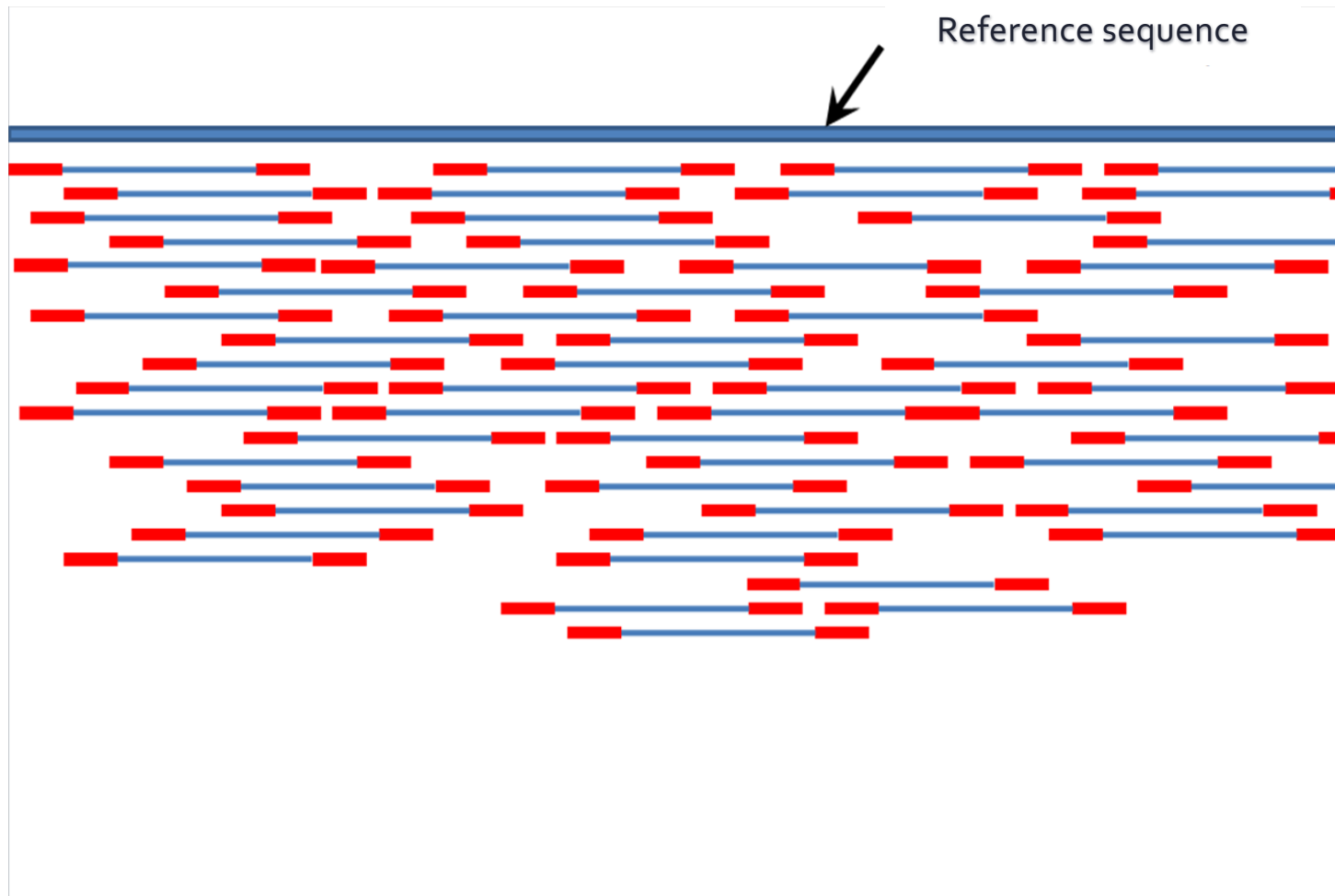


Next Generation Sequencing principle





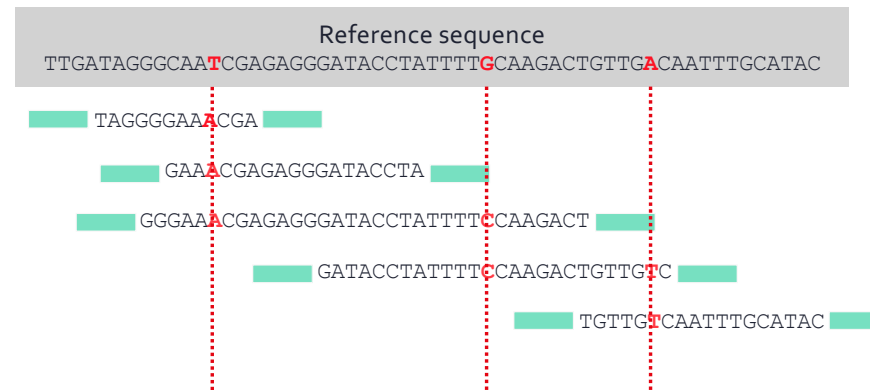
Next Generation Sequencing principle



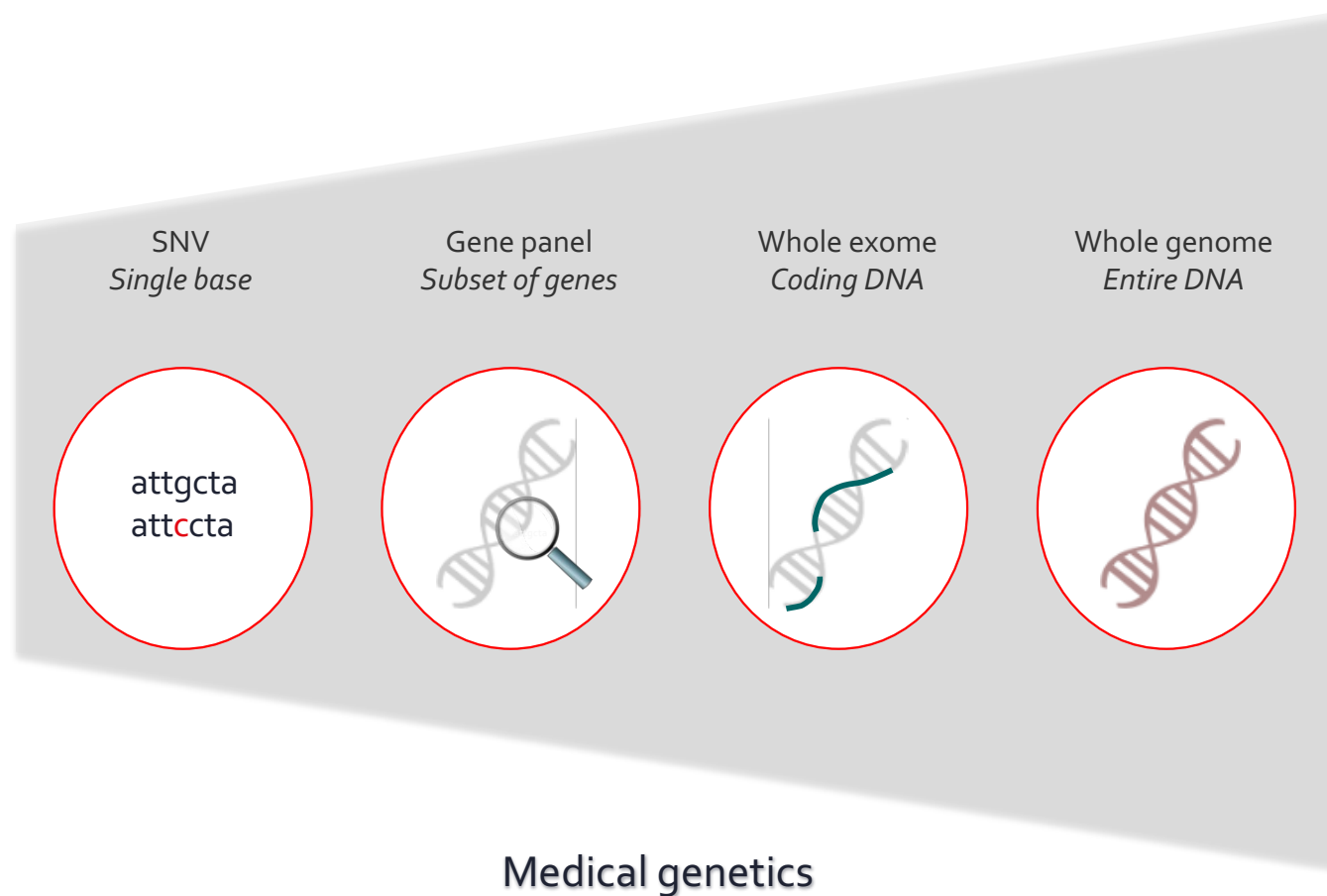


Examples of NGS clinical applications

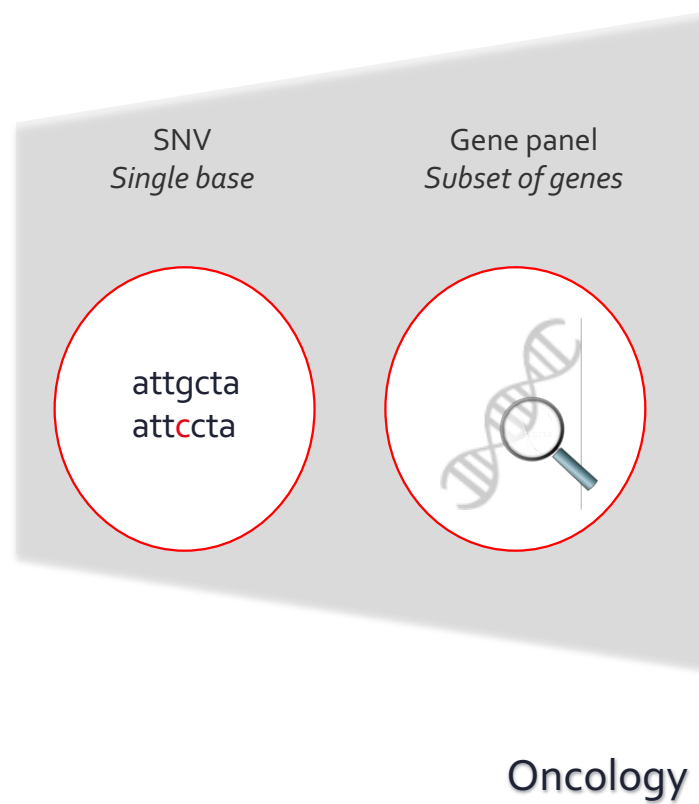
	Source DNA	Reference DNA
Oncology	Patient tumor or blood	Consensus human genome Patient germline
Microbiology	Patient	Pathogens genomes, resistance genes
Medical genetics	Patient	Family members, known defects
Pharmacogenetics	Patient	Drug-response or -sensitivity mutations

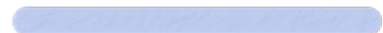


A matter of scale



A matter of scale



-  Results turn around time 
-  Limited reimbursement 
-  Clinically-actionable variants 
-  Incidental findings management 

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NGS in cancer diagnosis

- ❖ Identify **single nucleotide variants (SNVs)**, **insertions-deletions (indels)** to inform clinical management

at**t**cgggtcatgccccatagggg

Single Nucleotide
Variant (SNV)

at**g**cgggtcatgccccatagggg

Insertion

at**g**cgggtcat**cgtgtcc**gccccatagggg

Deletion

at**g**cgggtcatcgtgtccg....tagggg

ccca



Overview of a NGS bioinformatics pipeline

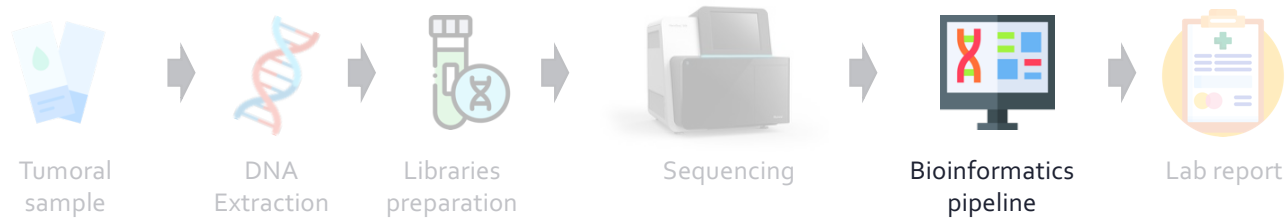


❖ Gene panels analysis in clinical routine

- Identify **differences**
- Identify **artifacts**: quality control
- Identify **somatic** vs. germline variants
- Variant **annotation**: does it provide clinically-useful information?



Overview of a NGS bioinformatics pipeline



Reads
filtering

Quality
control



Out of the sequencer: FASTQ file

```
Identifier —● @SRR566546.970 HWUSI-EAS1673_11067_FC7070M:4:1:2299:1109 length=50
Sequence —● TTGCCTGCCTATCATTTTAGTGCCTGTGAGGTGGAGATGTGAGGATCAGT
'+' sign —● +
Quality scores —● hhhhhhhhhhhghhghhhhhfhhhhhfffffe'ee['X]b[d[ed'[Y[~Y
Identifier —● @SRR566546.971 HWUSI-EAS1673_11067_FC7070M:4:1:2374:1108 length=50
Sequence —● GATTTGTATGAAAGTATACAACTAAACTGCAGGTGGATCAGAGTAAGTC
'+' sign —● +
Quality scores —● hhhhghfhhcgghghggfcffdhfehhhhcehdchhdhahehffffde'bVd
```

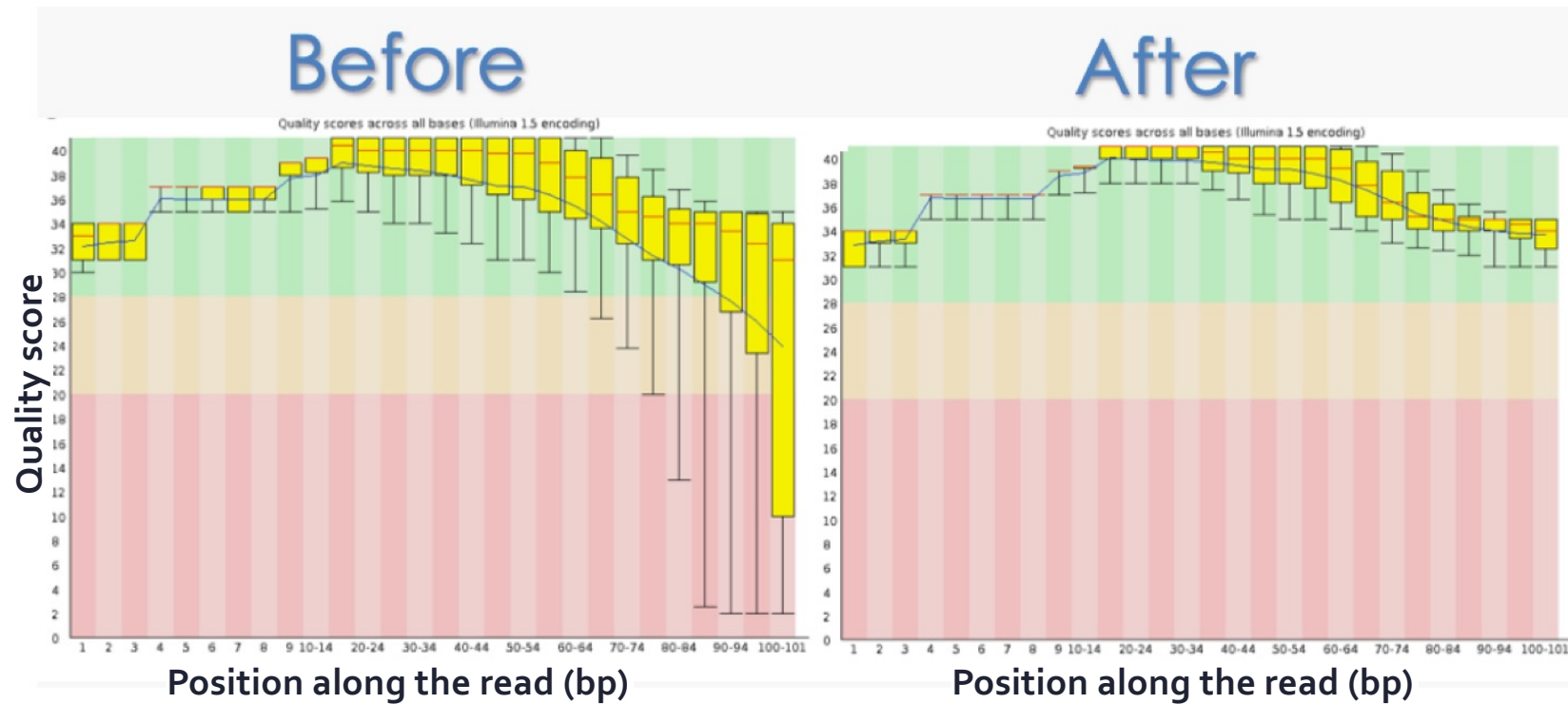
Each nucleotide has a **quality score (Phred score)**
representing the probability that a base was miscalled by the sequencer

$$Q = -10 \log_{10} P$$

Phred Score	Prob. of incorrect base call	Base call accuracy	Code
10	1 in 10	90%	J
20	1 in 100	99%	T
30	1 in 1'000	99.9%	^
40	1 in 10'000	99.99%	h

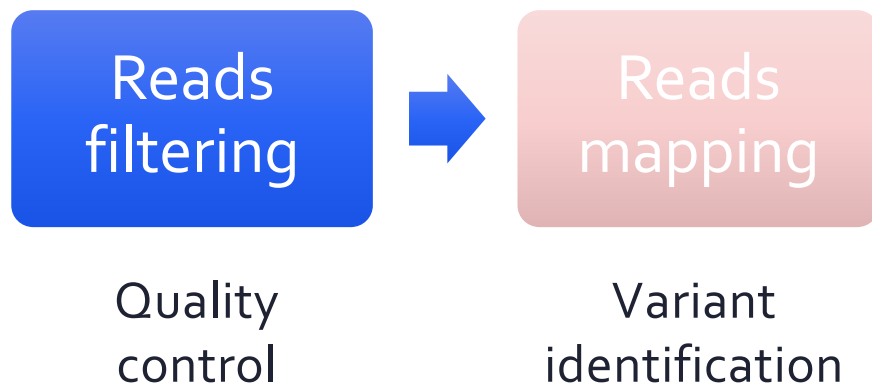
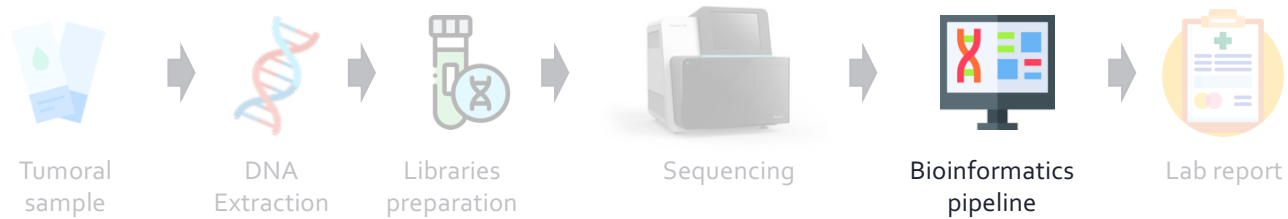


Quality-based reads trimming





Overview of a NGS bioinformatics pipeline



Let's align the reads

Reference genome

TCGCGCACAAG

Reference genome

CGTGGGACGAG

Reference genome

TCGCGCACAAGACGTGGGACGAG

» Short reads

- Likely to map at several positions along the reference genome

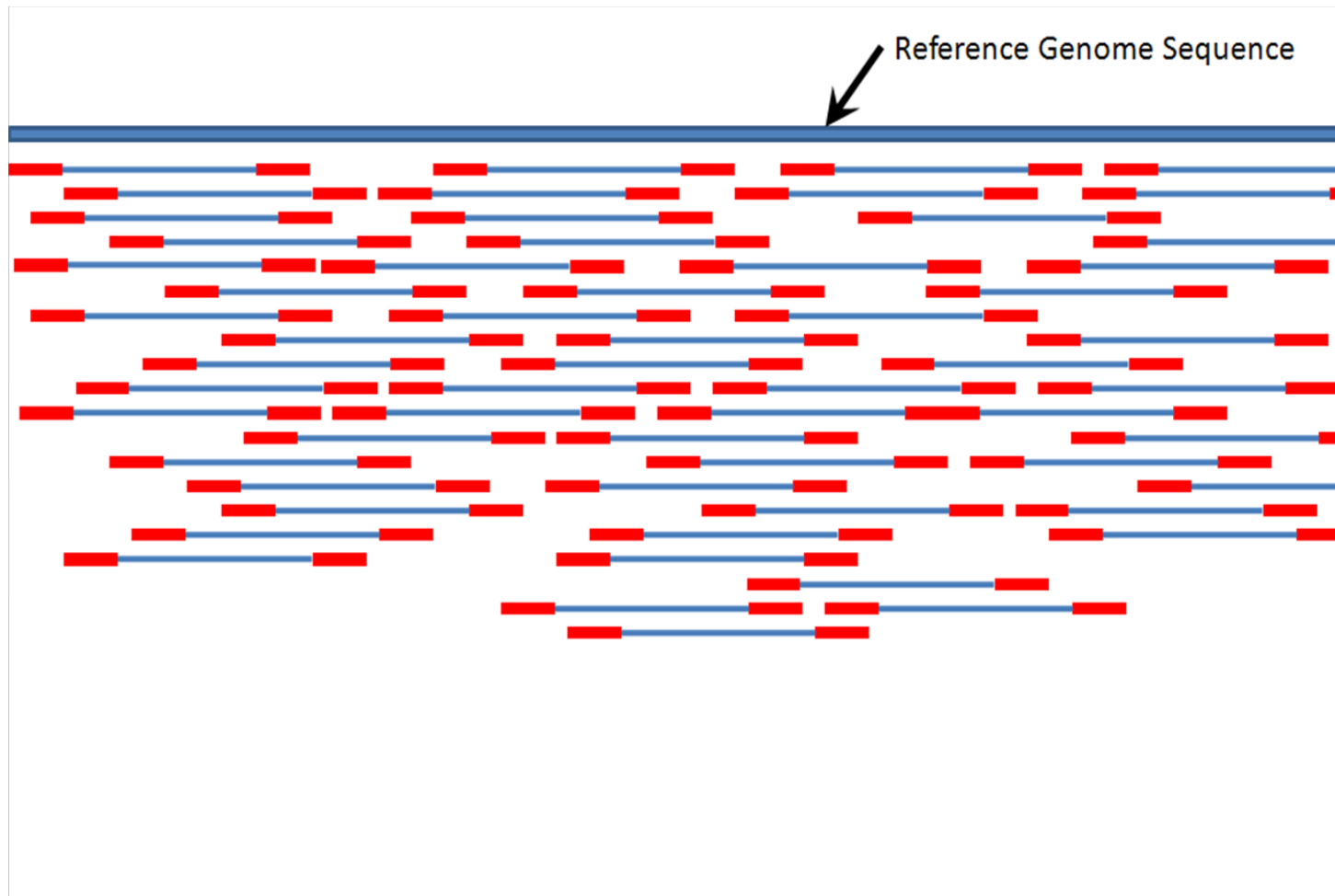
» Mismatches and gaps allowed

- Algorithms use scoring functions

» Longer reads

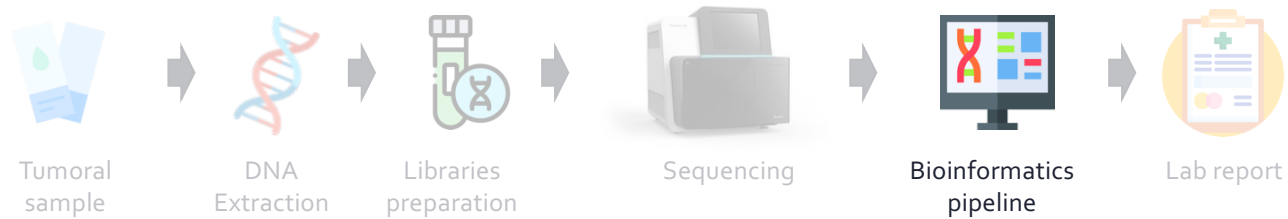
- Less ambiguous
- But computationally more expensive

Mapping: finding the best position for each read



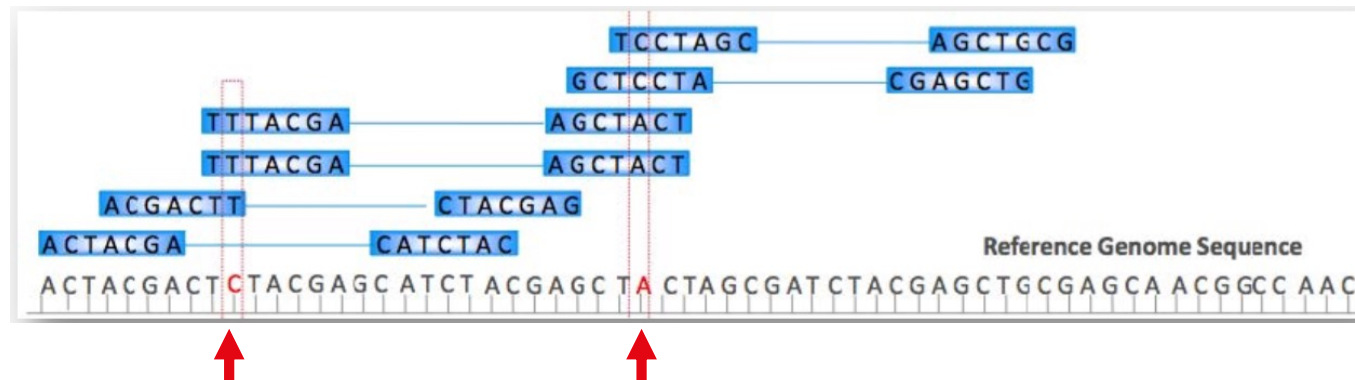


Overview of a NGS bioinformatics pipeline





Variant calling: putting it all together



True variant or technical error?

- ❖ Performed by the sequencer software or the bioinformatician
- ❖ Germline vs somatic calling
 - Germline: constitutional genome analysis, where variants occur in **50%** (heterozygous) or **100%** (homozygous) of the reads.
 - Somatic: no ploidy assumption, low frequency alleles.



VCF: Variant Call Format

```
##fileformat=VCFv4.1
##fileDate=20090805
##tcgaversion=1.1
##vcfProcessLog=<InputVCF=<file1.vcf>,InputVCFSource=<caller1>,InputVCFVer=<1.0>,InputVCFParam=<a1,b>,InputVCFgeneAnno=<anno1.gaf>>
##reference=ftp://ftp.ncbi.nih.gov/genbank/genomes/Eukaryotes/vertebrates_mammals/Homo_sapiens/GRCh37/special_requests/GRCh37-lite.fa
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">

##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">

##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">

##SAMPLE=<ID=NORMAL,Individual=TCGA-01-1000,File=TCGA-01-1000-1.bam,Platform=Illumina,Source=dbGAP,Accession=1234>
##SAMPLE=<ID=TUMOR,Individual=TCGA-01-1000,File=TCGA-01-1000-2.bam,Platform=Illumina,Source=dbGAP,Accession=4567>
##PEDIGREE=<Name_0=TUMOR,Name_1=NORMAL>
```

Optional: FORMAT field specifying data type
+ Per-sample genotype data

Fixed fields								Optional: FORMAT field specifying data type + Per-sample genotype data		
#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	NORMAL	TUMOR
20	14370	rs6054257	G	A	29	PASS	NS=3;DP=14;AF=0.5;DB;H2	GT:GQ:DP:HQ	0/0:48:1:51:51	1/0:48:0:51:51
20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017	GT:GQ:DP:HQ	0/0:49:3:58:50	0/1:3:5:65:3
20	1110696	rs6040355	A	G,T	67	PASS	NS=2;DP=10;AF=0.333,0.667;DB	GT:GQ:DP:HQ	1/2:21:6:23:27	2/1:2:0:18:2
20	1230237	.	T	.	47	PASS	NS=3;DP=13;AA=T	GT:GQ:DP:HQ	0/0:54:7:56:60	0/0:48:4:51:51
20	1234567	mtrosat1	GTC	G,GTCTC	50	PASS	NS=3;DP=9;AA=G	GT:GQ:DP	0/1:35:4	0/2:17:2



Output of the variant caller: VCF

VCF: Variant Call Format

HEADER	<pre>##fileformat=VCFv4.1 ##fileDate=20090805 ##tcgaversion=1.1 ##vcfProcessLog=<InputVCF=<file1.vcf>,InputVCFSource=<caller1>,InputVCFVer=<1.0>,InputVCFParam=<a1,b>,InputVCFgeneAnno=<anno1.gaf>> ##reference=ftp://ftp.ncbi.nih.gov/genbank/genomes/Eukaryotes/vertebrates_mammals/Homo_sapiens/GRCh37/special_requests/GRCh37-lite.fa ##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x> ##phasing=partial ##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data"> ##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth"> ##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency"> ##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele"> ##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129"> ##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership"> ##FILTER=<ID=q10,Description="Quality below 10"> ##FILTER=<ID=s50,Description="Less than 50% of samples have data"> ##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"> ##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality"> ##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth"> ##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality"> ##SAMPLE=<ID=NORMAL,Individual=TCGA-01-1000,File=TCGA-01-1000-1.bam,Platform=Illumina,Source=dbGAP,Accession=1234> ##SAMPLE=<ID=TUMOR,Individual=TCGA-01-1000,File=TCGA-01-1000-2.bam,Platform=Illumina,Source=dbGAP,Accession=4567> ##PEDIGREE=<Name_0=TUMOR,Name_1=NORMAL></pre>									
	Optional: FORMAT field specifying data type + Per-sample genotype data									
	Fixed fields								FORMAT	
	#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	NORMAL	TUMOR
BODY	20	14370	rs6054257	G	A	29	PASS	NS=3;DP=14;AF=0.5;DB;H2	GT:GQ:DP:HQ	0 0:48:1:51,51
	20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017	GT:GQ:DP:HQ	0 0:49:3:58,50
	20	1110696	rs6040355	A	G,T	67	PASS	NS=2;DP=10;AF=0.333,0.667;DB	GT:GQ:DP:HQ	1 2:21:6:23,27
	20	1230237	.	T	.	47	PASS	NS=3;DP=13;AA=T	GT:GQ:DP:HQ	0 0:54:7:56,60
	20	1234567	microsat1	GTC	G,GTCTC	50	PASS	NS=3;DP=9;AA=G	GT:GQ:DP	0 1:35:4

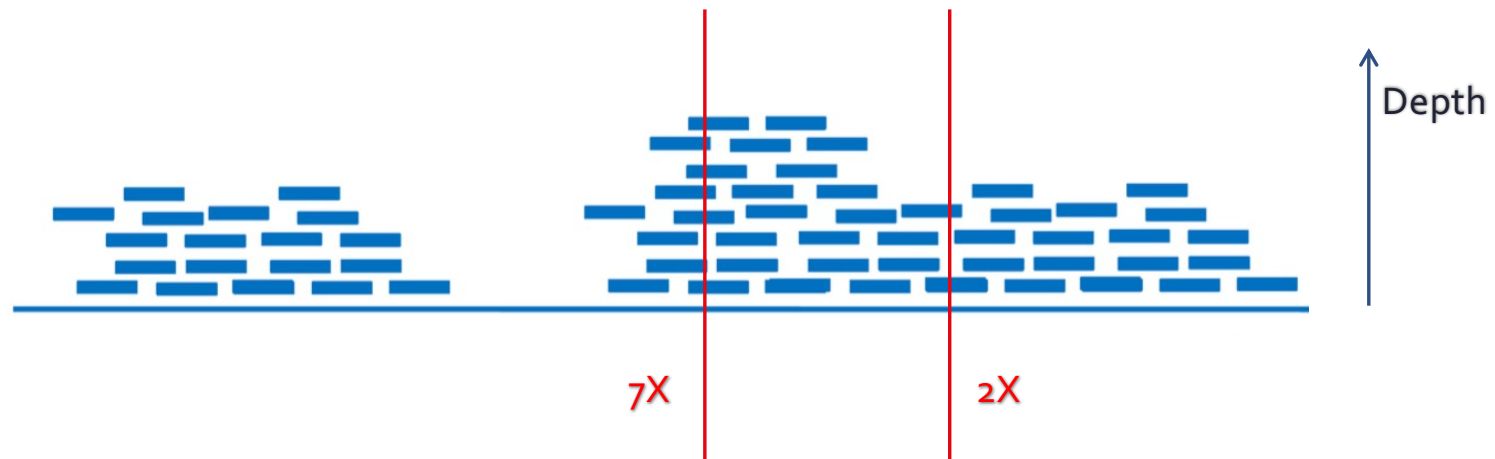




Things to watch out when assessing variant quality

Depth

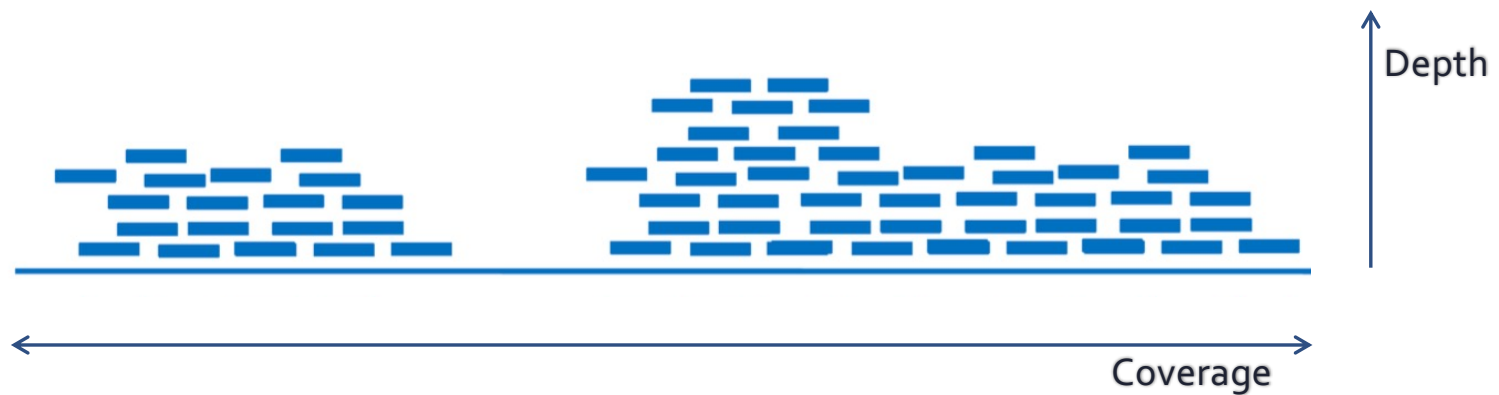
Depth: nb of reads that include a given nucleotide, at a given position



- ❖ Diagnosis: gene panel at 1500X, whole exome at 100X
- ❖ In oncology, impossible to detect low frequency clones with exome analyses

Coverage

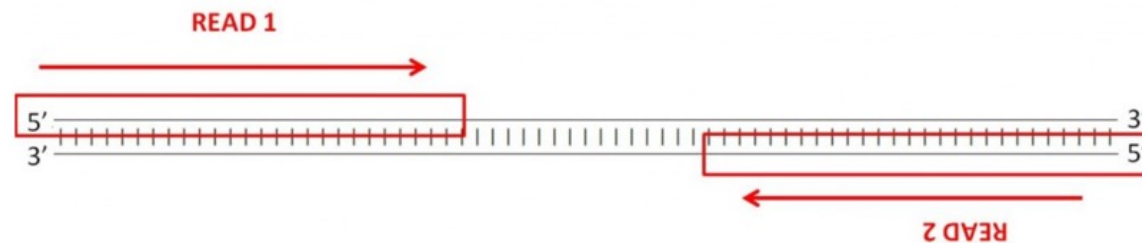
Coverage: % or nb of bases of a reference genome that are covered with a certain depth, e.g. 90% at 5X



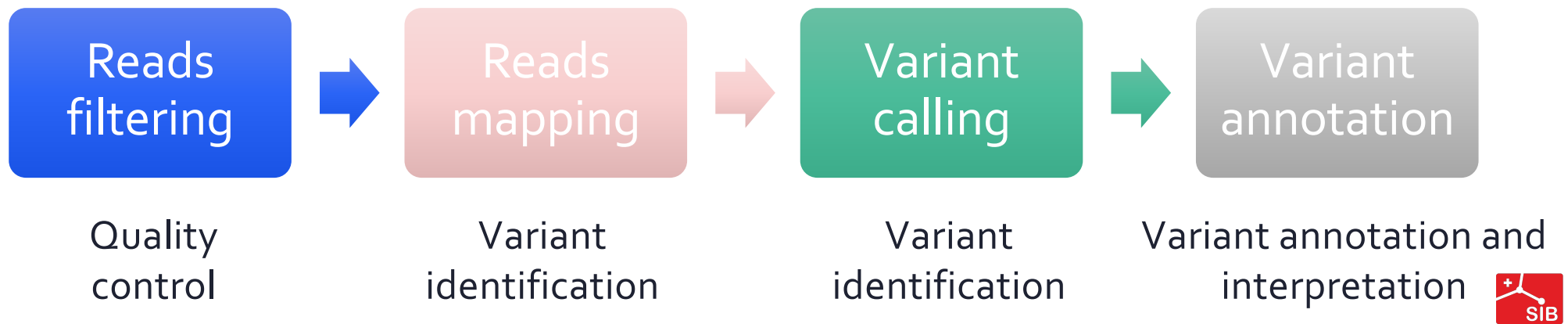
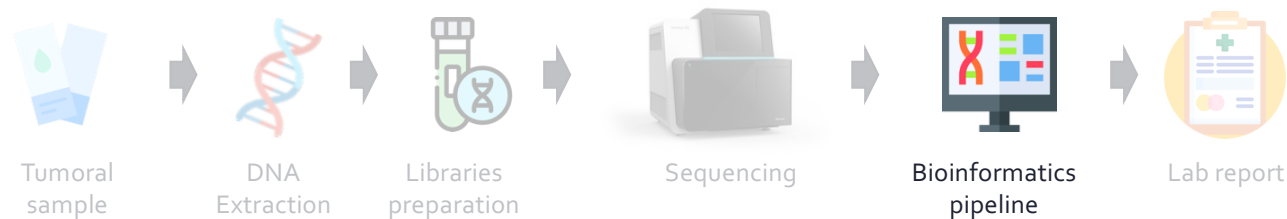


Strand bias in paired-end sequencing

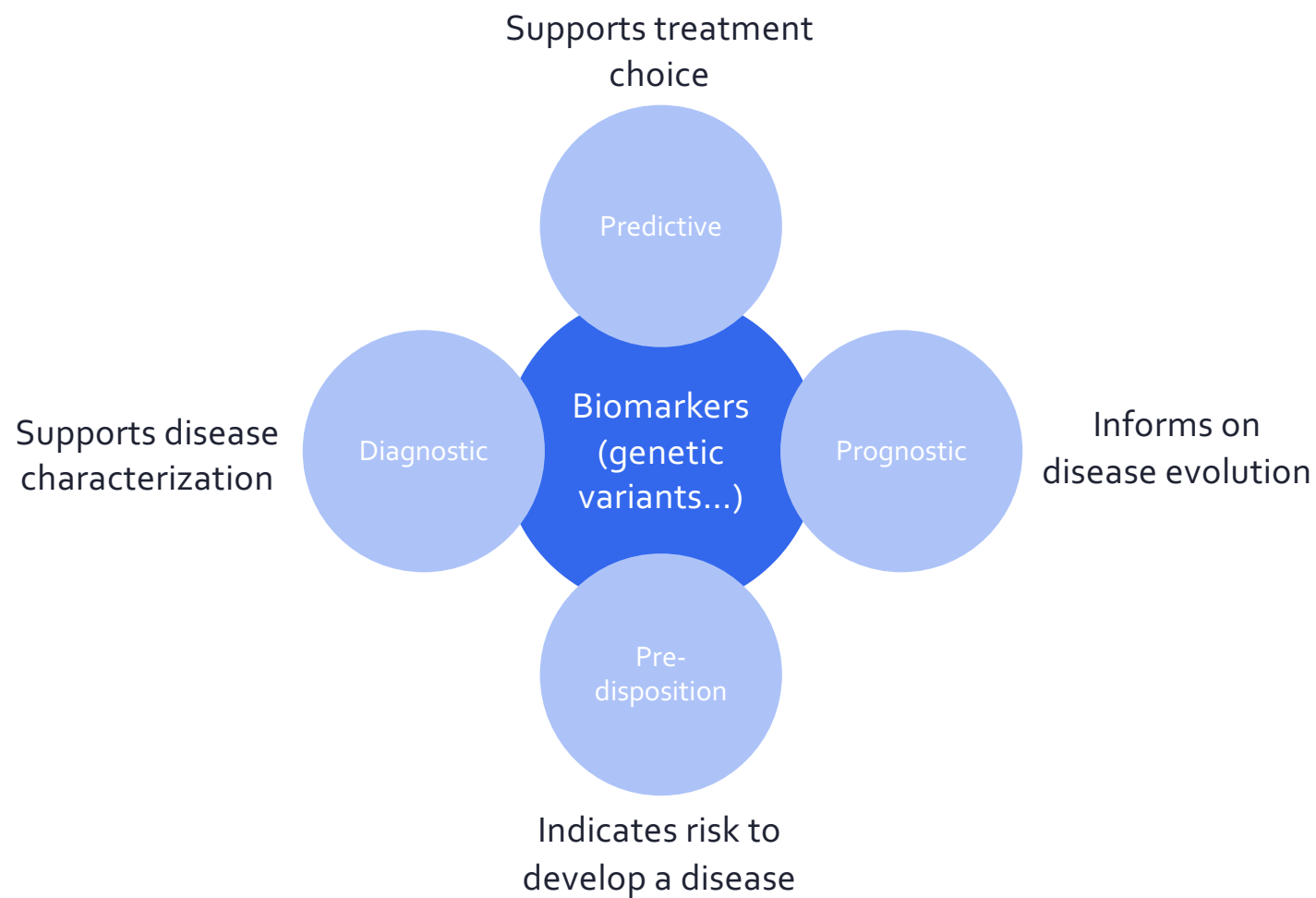
- ❖ Both DNA strands are generally sequenced
- ❖ Real mutations should occur on both with equal frequencies



Overview of a NGS bioinformatics pipeline



Personalized medicine





Medical genetics: focus on pathogenicity

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ACMG STANDARDS AND GUIDELINES

Genetics
inMedicine

Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology

Sue Richards, PhD¹, Nazneen Aziz, PhD^{2,16}, Sherri Bale, PhD³, David Bick, MD⁴, Soma Das, PhD⁵, Julie Gastier-Foster, PhD^{6,7,8}, Wayne W. Grody, MD, PhD^{9,10,11}, Madhuri Hegde, PhD¹², Elaine Lyon, PhD¹³, Elaine Spector, PhD¹⁴, Karl Voelkerding, MD¹³ and Heidi L. Rehm, PhD¹⁵; on behalf of the ACMG Laboratory Quality Assurance Committee

GENETICS in MEDICINE | Volume 17 | Number 5 | May 2015

Find **pathogenic** variants

i.e. genetic alterations increasing an individual's susceptibility or predisposition to a certain disorder





Oncology: focus on clinical significance

The Journal of Molecular Diagnostics, Vol. 19, No. 1, January 2017



SPECIAL ARTICLE

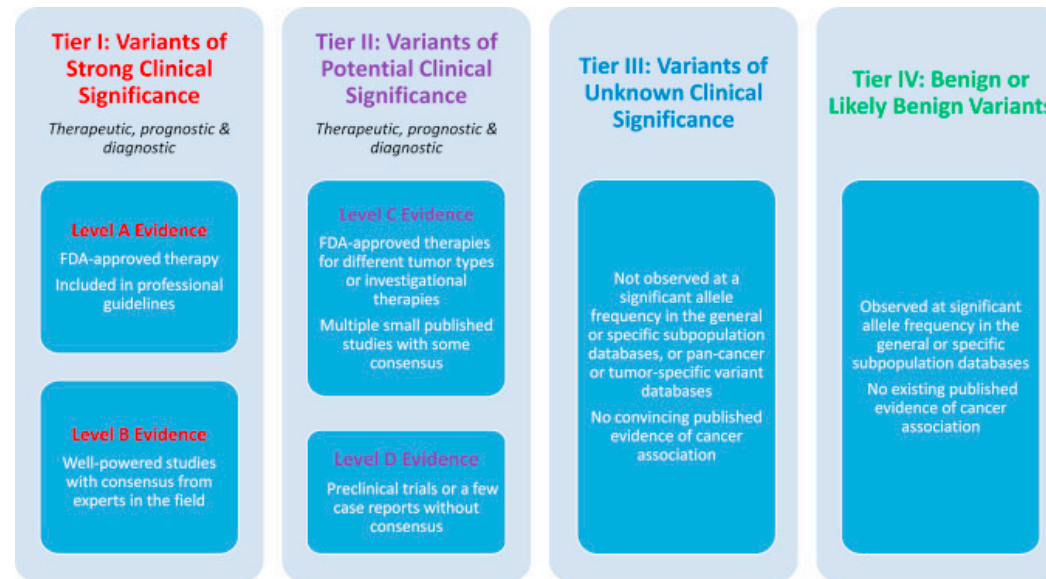
Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer



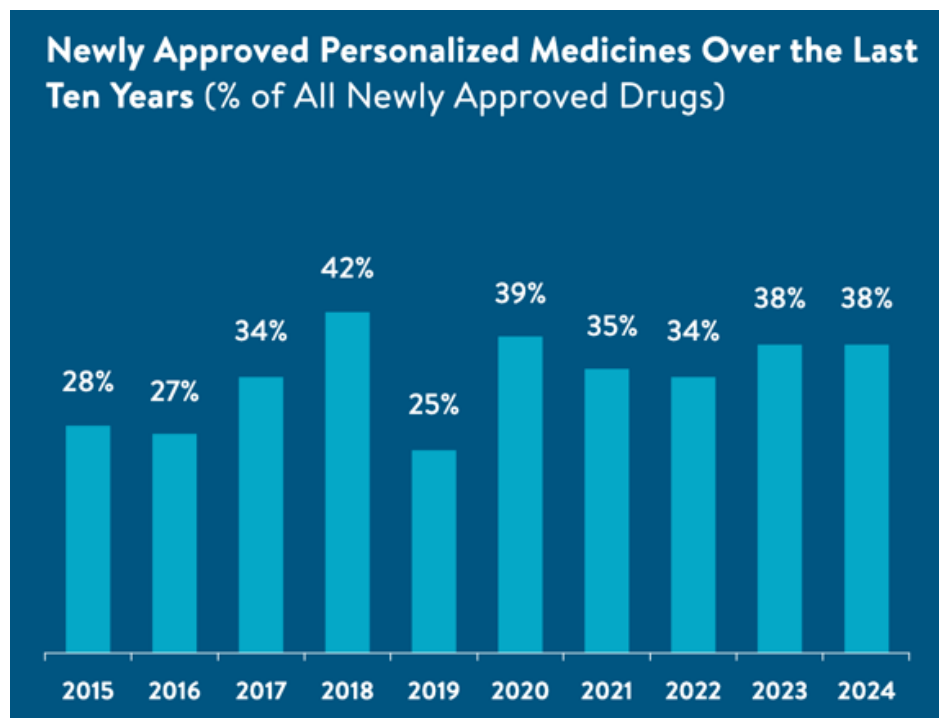
A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists

Find **actionable** variants

i.e. genetic alterations possibly having an impact on clinical care

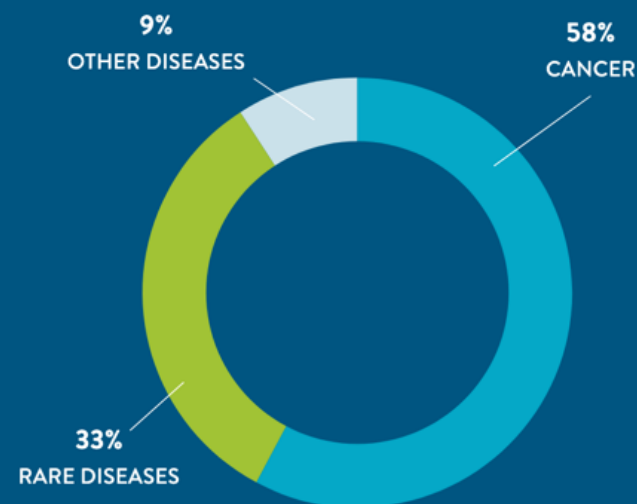


Personalized medicine and drug development



Newly Approved Personalized Medicines Are Indicated for Treatment of Rare Diseases, Cancers, and Common Diseases

Includes 18 New Molecular Entities and Six Newly Approved Gene or Cell-Based Therapies



Bioinformatics to guide the interpretation



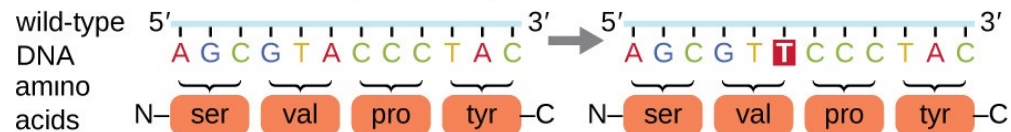
- ❖❖ Location of the variant (e.g. intron, exon, regulatory region, splicing site ...)
- ❖❖ Genes and transcripts affected by the variant
- ❖❖ Predict the variant effect on the protein sequence (e.g. stop gained, missense...)



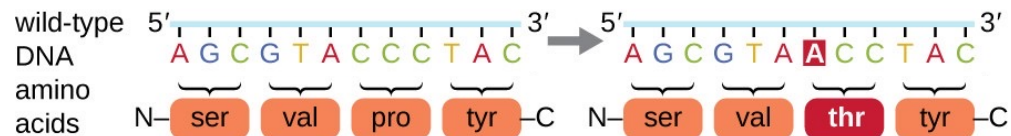
Predicting variants effect on the protein

Point mutations
(single base
substitution)

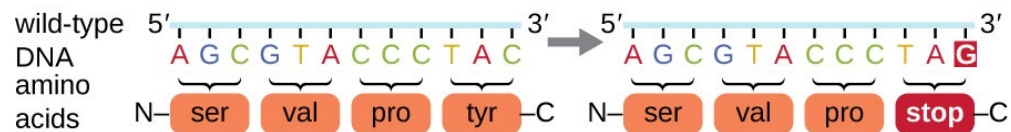
silent: has no effect on the protein sequence



missense: results in an amino acid substitution

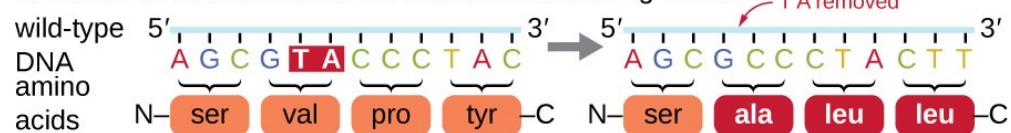


nonsense: substitutes a stop codon for an amino acid



Frameshift mutations
(insertion or deletion of
one or several bases)

Insertion or deletion results in a shift in the reading frame.



Bioinformatics to guide the interpretation



- ❖❖ Location of the variant (e.g. intron, exon, regulatory region, splicing site ...)
- ❖❖ Genes and transcripts affected by the variant
- ❖❖ Predict the variant effect on the protein sequence (e.g. stop gained, missense...)
- ❖❖ Predict the variant impact on the protein function

Predicting variants impact: examples of tools

TOOLS	SnpEff (ClinEff)	VEP	SIFT	PolyPhen-2	FATHMM
Variant effect and location (sequence ontology)	✓	✓			
Prediction of impact (score or category)	✓	←	✓ ←	✓	✓
Features used for impact prediction	Rules based on variant effect (stop gained, lost...)		AA conservation in related seq.	AA conservation and structural features	AA conservation and protein tolerance to mutations

© American College of Medical Genetics and Genomics **ACMG STANDARDS AND GUIDELINES** | Genetics
inMedicine

Standards and guidelines for the interpretation of sequence
variants: a joint consensus recommendation of the American
College of Medical Genetics and Genomics and the
Association for Molecular Pathology

Sue Richards, PhD¹, Nazneen Aziz, PhD^{2,3*}, Sherri Bale, PhD¹, David Bick, MD¹, Soma Das, PhD⁴,
Julie Gastier-Foster, PhD^{1,5}, Wayne W. Grody, MD, PhD^{6,7,8,9}, Madhuri Hegde, PhD¹⁰,
Elaine Lyon, PhD¹¹, Elaine Spector, PhD¹², Karl Voelkerding, MD¹³ and Heidi L. Rehm, PhD¹⁴,
on behalf of the ACMG Laboratory Quality Assurance Committee

GENETICS in MEDICINE | Volume 17 | Number 5 | May 2015

*Use a combination of tools and keep
variants with consensus prediction.*

(not exhaustive)

Bioinformatics to guide the interpretation



- ❖❖ Location of the variant (e.g. intron, exon, regulatory region, splicing site ...)
- ❖❖ Genes and transcripts affected by the variant
- ❖❖ Predict the variant effect on the protein sequence (e.g. stop gained, missense...)
- ❖❖ Predict the variant impact on the protein function
- ❖❖ Retrieve annotations from public databases

Knowledge bases

Cancer



OncoKB
COSMIC
CIViC
ClinVar
IARC TP53 Database

Genetic disorders



OMIM®
DECIPHER
orphanet

Constitutional variants



dbSNP
Short Genetic Variations
ClinVar
HGMD

Population



gnomAD

Repositories



UniProt
RefSeq
Ensembl
PubMed

Non exhaustive



- KEEVQLLVFGLTANSSPSVOCRSPRGKNIQGKTLVLSQLELQDSGTWTCVTLQNQKVEFKIDIVVLAFO
KEEVQLLVFGLTANSSPSVOCRSPRGKNIQGKTLVLSQLELQDSGTWTCVTLQNQKVEFKIDIVVLAFO
KEEVQLLVFGLTANSSPSVOCRSPRGKNIQGKTLVLSQLELQDSGTWTCVTLQNQKVEFKIDIVVLAFO
KEEVELLVFGLTANSSPSVKCRSPGGKNIQGRTLVPQLERODSGTWTCTVSODQTVEFKIDIVVLAFO
KEEVELLVFGLTANSSPSVKCRSPGGKNIQGRTLVPQLERODSGTWTCTVSODQTVEFKIDIMVLAFO
KEEVQLLVFGLTANSSPSVOCRSPTGKNIQGKTLVLSQLELQDSGTWTCVTLODQKVEFKIDIVVLAFO
KEEVLQVFGLTASSPSVECTSPRGKRIQMKTLYSQLEIQDSGTWKCCTVSQHPPELLFKINVVVLAFO
KEEVLQVFGLTANSSPSVECMSPRGKRIIRMKTLMSQLEIQDSGTWKCCTVSQHLELVFKINIVVLAFO
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KAEVELWVFKVTVSAKPSFCKGPRNGVATS-KVVPNLRIQDSGTWTCSTVTONKETVKINIVVLGFK
KEEVELWVFRVTFNSDPPIECKHKSSNIVKSKAFTHSLRIQDSGIWNCCTVTLNQKHSFDMKLSVLGFA
KEEVELWVFKVTFSSNPLETECKHKKGKVYSSKVLMSNLRVQDSDFWNCCTVTLNQKNWFGMTLSVLGFA
- Conservation

Identifying a damaging mutation: is it always bad?

- ❖ Keep the mutation in context: what is the gene function?
 - **Tumor suppressor gene**
Damaging mutations are pathogenic.
 - **Oncogene**
Activating mutations are pathogenic.
(beware: damaging mutation can be activating!)

**Keep the gene function in mind
when interpreting its deleteriousness**

Outline

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clinical bioinformatics

Why clinical bioinformatics?
*Next Generation Sequencing (NGS)
in medical diagnosis*

Overview of an oncology NGS
diagnostic pipeline

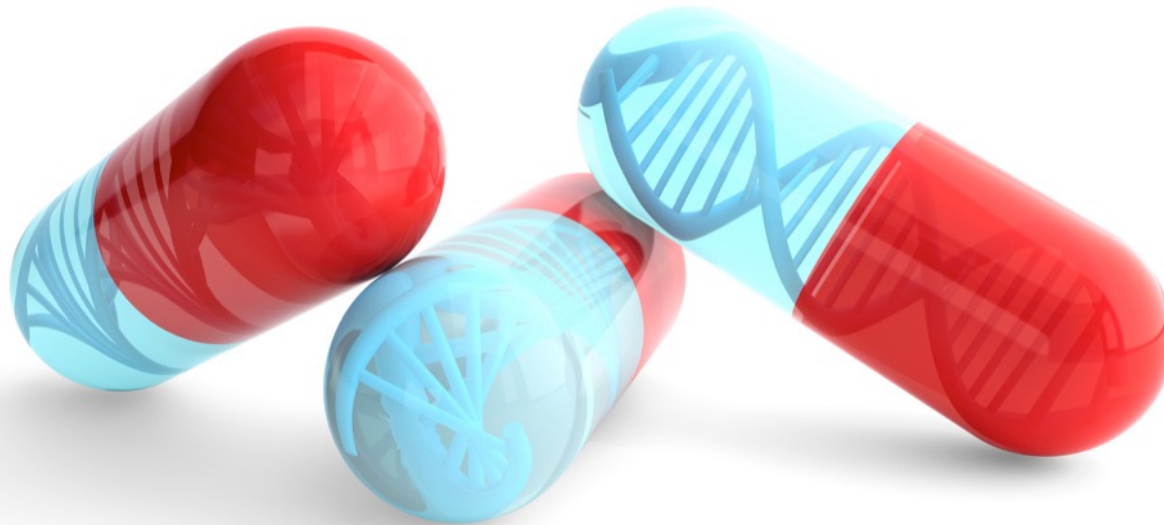
Other considerations
in clinical settings

Real-life constraints in the clinics



Certificate of Advanced Studies (CAS) in Personalized molecular oncology

pmo.unibas.ch



CAS PMO: 4 modules and a mini-thesis



Tumor biology & Genetics
4 days

Lausanne

- Cytogenetics and molecular genetics
- Genetic modifications
- Tumor biology: solid and hematological
- Tumor genetics



Molecular Pathology
4 days

Basel

- Omics technologies
- From sample to data: extraction, sequencing, panels
- Quality control and accreditation
- Molecular profile interpretation, reports



Clinical Bioinformatics
4 days

Lausanne

- NGS data processing: mapping, calling, annotation
- Data quality control
- Hardware, security, privacy
- Artificial intelligence applications



Clinical Oncology
4 days

Basel

- Tumor physiology & immunology
 - Prognostic and predictive markers
 - Interpretation of genetic results
 - Clinical trials and tumor board
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